



YOGITA Wilson Karke
PhD Research Scholar, IIIT, Bangalore

Contact



+91 9902020338



[linkedin.com/in/yogitawilson](https://www.linkedin.com/in/yogitawilson)



@YogitaWilson



yogita.pimpalkar@iiitb.ac.in

Skills

- C, C++
- Python,
- IPsec, IKE, IKE2
- Cryptographic algorithms
- TCP/IP,
- SDWAN
- 5G/6G

Research Experience

PhD Research Scholar, IIIT, Bangalore (1st Aug 2025 – present)

My research focuses on IoT networks, network slicing, optimization techniques for network resource allocation, AI-driven security frameworks, Post-Quantum Cryptography, and emerging 6G technologies.

Industrial Experience

- **Senior software engineer at Cisco Systems**
(9th Aug 2017 – 31st Sep 2021)
Development, Automation, Performance testing, and Support for various IPsec and SDWAN projects.
- **Tech Lead at Infosys India Ltd**
(28th Jul 2014 – 31st Jul 2017)
Lead a team of three and managed the work for the IPsec project.
- **Senior software engineer at Tech Mahindra Ltd**
(22nd Sept 2004 -16th Nov 2009)
Lead the team of four , designed and developed different Alcatel SSG platform features and enhancements.

Education

Post-Graduation (CGPA :3.35)	MS by Research (2022-present) International Institute of Information Technology-Bangalore
Post-Graduation Diploma (Distinction)	PGDBA Finance (2012-2014) Symbiosis Institute, Pune.
Graduation (71.33% Distinction)	BE Information Technology (2000- 2004) P.E.S. College OF Engineering, Maharashtra

Key Industry Projects

SD-WAN Performance Automation

- Designed Python automation frameworks to benchmark throughput across SD-WAN features including AppQoE, AppNav, DRE, SSL Proxy, and Unified Threat Defense (UTD).

Honors-Awards

- Received **'You Amaze'** Rewards for various projects at Cisco.
- **'Infosys Insta'** award in 2016 And 2017 for all-around contributions to Organization and Client projects.
- **'Pat On the Back'** award in 2009 for exceptional contribution to Gateway Screening Projects at TechMahindra.

- Automated performance validation across multiple Cisco platforms, improving test coverage and reducing manual performance testing.

SSL/TLS Proxy (Cisco SD-WAN Security)

- Developed management layer and telemetry modules for SSL/TLS Proxy enabling encrypted traffic inspection by Unified Threat Defense (UTD).
- Designed YANG data models for configuration and operational statistics.
- Implemented controller integration and statistics collection modules.
- Delivered stable feature through unit and integration testing across multiple deployments.

IPsec Data Plane Acceleration

- Enhanced IPsec control plane and data path components for improved security and performance.
- Stabilized ISM hardware accelerator cards and optimized IPsec switching for Cisco ISR-G2 platforms.

Dynamic Application Policy Routing (DAPR)

- Designed and implemented CLI configuration interfaces, infrastructure APIs, operational telemetry modules, and debugging tools.
- Enabled dynamic application-aware traffic steering based on link utilization and application policies to optimize WAN bandwidth.

Gateway Screening Range Expansion

- Implemented automation for combining and splitting point code ranges in Gateway Screening tables without service disruption.
- Owned end-to-end feature lifecycle: architecture design, development, testing, and customer release.

Origin-Based Routing

- Developed routing functionality allowing route selection based on both destination and originating user attributes.
- Implemented database preparation scripts and unit test frameworks to validate routing logic.

Publications

Peer-Reviewed Accepted International Conference Papers

[1] Y. Pimpalkar, S. Ravindran, J. Bapat and D. Das, "New Attacks on Cloud Container-Based IoT Systems in 5G-Advanced and Beyond," 2023 IEEE International Conference on Electronics, Computing and Communication Technologies (**CONECCT**), **Bangalore, India, 2023**, pp. 1-6, doi: 10.1109/CONECCT57959.2023.10234780.

Honors-Awards

- **'Best Track Award'** at IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT). IEEE, 2023.
- **'Best Paper Award'** at IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT). IEEE, 2023.

[2] Y. Pimpalkar, S. Ravindran, J. Bapat and D. Das, "A Novel E2E Path Selection Algorithm for Superior QoS and QoE for 6G Services", submitted manuscript to **IEEE Transactions on Network and Service Management**.

[3] S. Ravindran, **Y. Pimpalkar**, J. Bapat and D. Das, "Multi-User Service Throughput Estimation for Adaptive Resource Management in 5G-Advanced and Beyond Sliced Networks," 2023 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT), Bangalore, India, 2023, pp.1-6, doi:10.1109/CONECCT57959.2023.10234812.

[4] Y. Pimpalkar, S. Ravindran, J. Bapat and D. Das, "Implications of Generative AI on IoT Security: A Dual Perspective", in *Advances in Computing and Network Communications*, Cham, Switzerland: Springer, 2026, pp. 83–104, doi: 10.1007/978-3-032-05507-1_3.

Patent Application Numbers

[1] Patent Application No. 202441057669
TITLE: SYSTEM AND METHOD FOR DETECTING CLOUD CONTAINER-BASED ENERGY DEPLETION AND SYNERGISTIC POWER ATTACKS IN AN IOT NETWORK
Date of Filing: 30/07/2024

[2] Patent Application No. 202441057670
TITLE: SYSTEM AND METHOD FOR ESTIMATING AND OPTIMIZING THROUGHPUT IN O-RAN SLICING ENVIRONMENT
Date of Filing: 30/07/2024

[3] Patent Application No. 202541123553
TITLE: SERVICE-AWARE OPTIMAL END-TO-END (E2E) PATH SELECTION SYSTEM AND METHOD FOR SOFTWARE DEFINED NETWORKS
Date of Filing: 08/12/2025